HUMBER POLYTECHNIC | Faculty of Applied Sciences & Technology

ENGI 4000

CAPSTONE PROJECT 1

CHAPTER 3

IDENTIFYING NEEDS





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Executive Summary

This chapter identifies and organizes the customer's needs for the High-Speed Camera Robot. It is organized into four sections where Section 1 explains the team’s data collection approach and presents the raw interview data, the prompt questions used, the statements gathered from representative users, and the needs interpreted from them. Section 2 organizes the interpreted needs into a hierarchy of grouped final customer need statements to properly identify what exactly is being desired from the project or what is being evaluated. Section 3 establishes the relative importance of each need statement and explains the approach used to weigh them. It also delves into why each of the needs were rated as seen in the results. Lastly, Section 4 reflects the results and the process and presents a schematic of what need is being desired mainly as gathered from the previous sections.

Section 1: Identifying Customer Needs

1.1 Data Collection Approach

In terms of data collection, there will be a focus on conducting this through one-on-one interviews, and surveys with semi-structured interviews as the primary method. This will allow each interview to be scheduled individually while getting detailed perspectives on the feasibility of the project.

The interview will consist of subjects that will associate with the primary and secondary target markets being independent and small-studio film and commercial production, VFX compositing, stop-motion and time-lapse creators, product and tabletop photography. Each interview is planned, with each structured around the prompt questions, with a team member working to ask questions and record statements of their answers. As for persons that cannot make time or unable to get to location to conduct the interviews, a survey would be provided with the most important questions needed to be answered regarding this project.

1.2 Step 1: Interview Question Prompts

Prompt questions are as follows:

- Q1. Can you tell us about a recent activity where you needed the camera to move in a complex or repeated way?
- Q2. How do you currently achieve moving camera shots that need to be fast or repeated exactly the same way?
- Q3. What challenges do you experience when you try to get a camera to move to look smooth, especially at high speed?
- Q5. What do you do use when you need a movement that is too difficult to do by hand?
- Q6. If you could design a perfect tool for your camera system to work, what would it need to do?
- Q7. How would an automated camera rig fit into your future projects or activities?
- Q8. What would make you trust an automated camera rig around your set, your equipment, or your team?
- Q9. What concerns would you have about a robotic camera system moving near people or expensive equipment?
- Q10. What budget range would be realistic for a tool like this in your projects or activities?
- Q11. How much time would be reasonable to spend setting up and programming a shot before a shot?

1.3 Steps 2 and 3: Raw Data and Interpreted Needs

The table below presents the customer statements collected during the interview with Matthew Kennedy, Director of the Film department at Humber Polytechnic, along with the interpreted needs developed from his responses. The interview focused on the challenges of filming product advertisements as shown below

Matthew Kennedy– Director of Film department at Humber Polytechnic

Question prompt	Customer statement	Interpreted need
Q1	On a recent project, we filmed products for advertisements which require the camera to be moving in certain angles and directions with the appropriate speed which can be complicated to maintain throughout the shoot.	The camera system repeats a programmed set of sequences of camera movement to capture scenes that the human body can't maneuver similarly.
Q2	For this, we take multiple shots of the same scene trying to mimic the movement as much as possible through hand movements on the camera.	The camera system follows the same desired path every time.
Q3	We experience shaky movements that messes the shots, and the movement is not flowing properly at high speeds.	The camera system accelerates and decelerates without visible vibration at high frame rates.
Q4	We use a handheld gyro stick that the camera is mounted on to improve stability.	The camera system maintains its desired orientation regardless of any type of movement.
Q5	Something where I can drag points on a timeline, hit preview, and watch the move happen on screen before it runs on set.	The operator can preview a programmed move before it runs on set.
Q6	If it could also fire the lights, the movement and motion track vision at the same instant, that would save the need to make separate triggers.	The camera system carries a single area that controls all its features.
Q7	In our classroom I'd want a hard stop I can hit fast if something looks wrong, especially with students around it or it is stopping on its own in the presence of someone in its way.	The camera system stops immediately when the operator presses an emergency stop or stops when a person's presence is detected in the way of its motion.

Q8	The robotic arms I've seen quoted is very costly, I'd need something that can be easy and affordable to be replaced especially if the arm is damaged.	The camera is made in a way that the parts can be replaced for easy fix.
Q9	The budget range realistic for this would be $5000.	The camera system is affordable.
Q10	We don't have a dedicated stage, so anything that takes more than twenty minutes to rig eats into the shoot day.	The system needs to be set up quickly in any space

Section 2: Organize the Needs into a Hierarchy

This section organizes the interpreted needs into groups based on similar ideas and requirements of the project. This develops a clearer customer need statement which helps identify the main requirements as shown below.

Table 2.1

Group #	Interpreted Needs (Step 3 statements)	Customer Need Statement (Step 4)
Group 1	The camera system follows the same desired path every time.	The camera system repeats programmed camera paths each time
Group 2	The camera system accelerates and decelerates without visible vibration at high frame rates.	The camera system provides smooth camera movement without sudden acceleration or deceleration.
Group 3	The camera system maintains its desired orientation regardless of any type of movement.	The camera system maintains the desired camera orientation during movement.
Group 4	The operator can preview a programmed move before it runs on set.	The camera system allows the operator to program and preview camera movements before execution.
Group 5	The camera system carries a single area that controls all its features.	The camera system provides centralized control of camera movement and connected equipment.
Group 6	The camera system stops immediately when the operator presses an emergency stop or stops when a person's presence is detected in the way of its motion.	The camera system provides immediate and effective safety stopping capabilities during operation.
Group 7	The camera is made in a way that the parts can be replaced for easy fix. Needs to be affordable overall.	The camera system uses affordable, replaceable components to simplify maintenance and repairs.
Group 8	The camera system needs to be set up quickly in any space	The camera system can be quickly set up and operated in different spaces without requiring a dedicated stage.
Group 9	The camera system performs movements with high accuracy	The camera system can perform programmed movements very

		accurately with little to no deviation from set path.

2.2 Comments and Observations

What was observed was that repeatability is an important requirement because the camera needs to follow the same programmed path for multiple shots. This helps make sure that each take is as similar as possible and reduces differences caused by manual camera movement.

Also, smooth movement and camera orientation are important for getting good quality footage. Sudden changes in speed or unwanted changes in the camera angle could cause shaking or affect the final shot, especially when filming at high frame rates, hence the need for stability.

Programming and previewing would make the system easier to use and easier to demonstrate for students who want to use it. The operator should be able to create a movement and see what it will do before running the actual movement. This can help prevent mistakes and reduce the need for the operator to have programming experience.

Centralized control would make the system easier to operate by allowing the different functions to be controlled from one place. This could also make it easier to synchronize the camera with other equipment used during a shot.

Safety is important because the system will be operating around people and equipment. Having an emergency stop and the ability to stop the system if something is detected in its path can help prevent injuries and equipment damage that could lead to greater problems in the long run.

Replaceable parts and affordability are also important because the project is intended to be a lower-cost alternative to more expensive robotic camera systems. Being able to replace individual components instead of the entire system could also make repairs easier and cheaper.

Quick setup is important because the system may be used in different classrooms, studios, or other spaces without a dedicated stage. The system should therefore be easy to set up, move, and pack away when it is not being used.

Overall, what is being looked at is the movement accuracy, motion quality, ease of use, control, safety, cost, maintenance, and setup.

Section 3: Establish the Relative Importance of the Needs

This section shows the relative importance of each customer's needs for the Camera system using a rating scale from 1 to 5, where 1 represents the lowest importance and 5 represents the highest importance. Each need is assigned a rating based on its impact on the system's performance, safety, usability, and cost as shown below.

Need	Customer Need	Importance (1–5)	Reason
1	The camera system repeats programmed camera paths each time.	**5**	Repeatability is one of the main reasons for using a robotic camera system instead of manually moving the camera, hence rated so high.
2	The camera system provides smooth camera movement without sudden acceleration or deceleration.	**4**	Smooth movement is necessary to produce usable footage, especially for high-speed and slow-motion shots. Would save a lot of time, energy and data space.
3	The camera system maintains the desired camera orientation during movement.	**3**	Maintaining orientation helps keep the subject framed correctly during programmed movements.
4	The camera system allows the operator to program and preview camera movements before execution.	**3**	This makes the system easier to use and helps prevent mistakes before running the actual movement but can be manually programmed without the need for a screen.
5	The camera system provides centralized control of camera movement and connected equipment.	**2**	Centralized control improves usability and synchronization of its functions, but the camera can still perform its main function without all external equipment being connected or used, which is why it is rated low.
6	The camera system provides immediate and effective safety stopping capabilities during operation.	**5**	Safety is critical because the system will operate around people and expensive equipment where damage to either is very costly.
7	The camera system uses affordable, replaceable components to simplify maintenance and repairs.	**2**	The project has a limited budget, and replaceable parts can reduce repair costs and downtime for the long run.
8	The camera system can be quickly set up and operated	**1**	The system is intended for classrooms and small studios, so portability and setup time

	in different spaces without requiring a dedicated stage.		are important but can be done at any time which rates it the lowest.
9	The camera system can perform programmed movements very accurately with little to no deviation from set path.	**4**	Accurate movements are necessary to produce desired scenes in footage as it contributes to saving time and energy trying to perfect the movement.

3.2 Relative Importance of Customer Needs

The highest ratings were given to Numbers 1 and 6, with a rating of 5, as repeatability and safety are directly related to the main purpose and safe operation of this project. Repeatability is important because the system needs to produce consistent camera movements, while safety is important because the system will operate around people and expensive equipment that can be costly to replace if accidents occur.

Numbers 2 and 9 were given a rating of 4 because smooth movement and accuracy have a strong effect on the quality of the footage produced. Smooth acceleration and deceleration help prevent unwanted vibrations, while accurate movements help produce the desired scenes without wasting time and energy trying to perfect the movement.

Numbers 3 and 4 were given a rating of 3 because maintaining camera orientation and being able to program and preview movements improves the usability of the system. However, the system can still perform its main function without these features, which is why they were given a moderate rating.

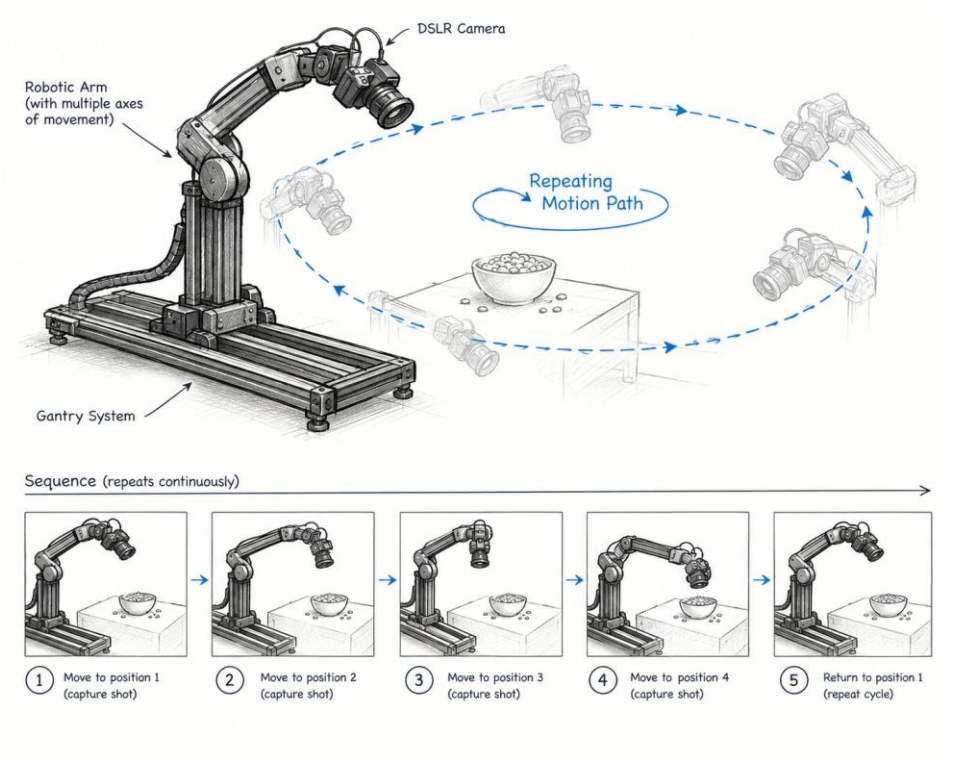
Numbers 5 and 7 were given a rating of 2 because centralized control and replaceable components are useful additions but are not essential at all to the main function of the system. Centralized control would make it easier to operate the camera alongside other equipment, while replaceable parts would simplify repairs and reduce downtime in the long run. However, the system can still perform its main camera movement functions without these features being a high priority.

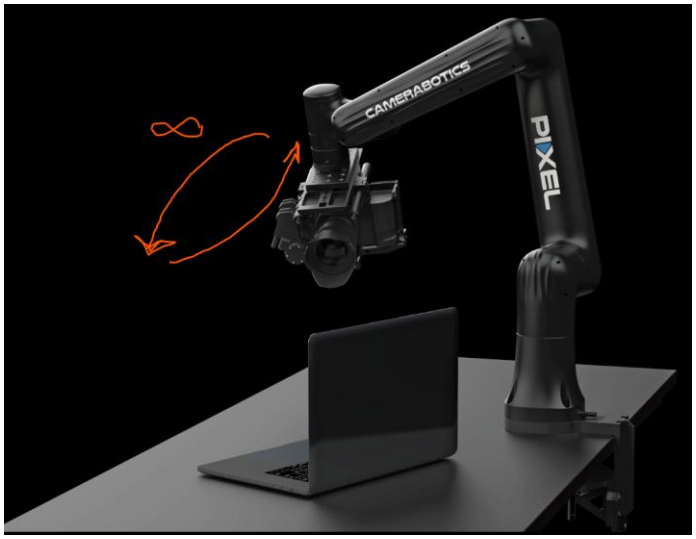
Lastly, Number 8 was given the lowest rating of 1 because portability and setup time are less important than the functionality of the system. Since the system is intended for classrooms and small studios, it will most likely be kept in one room for extended periods, meaning that quick setup in different spaces is not as important as repeatability, smooth movement, accuracy, and safety.

Section 4: Reflect on the Results and the Process

Number 1 - Repeatability of Camera system movement

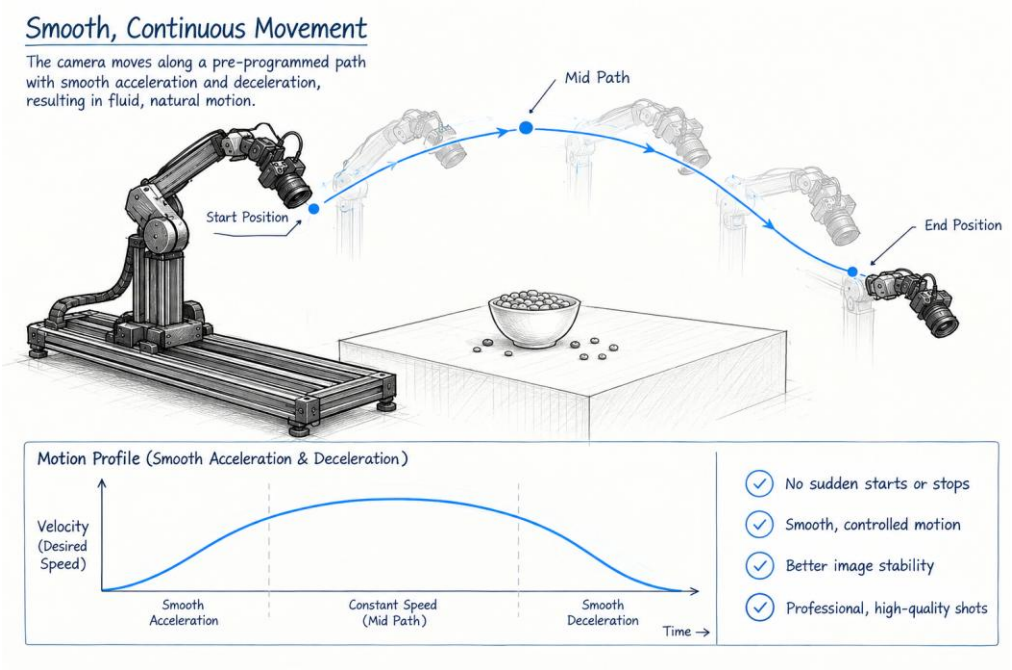
This is one of the main requirements of the system because the purpose of the system is to create camera movements that can be repeated consistently. Manually moving a camera makes it difficult to follow the exact same path for multiple takes. The use of motors, position feedback, and programmed movements can directly address this need by allowing the system to record and repeat a specific camera path.





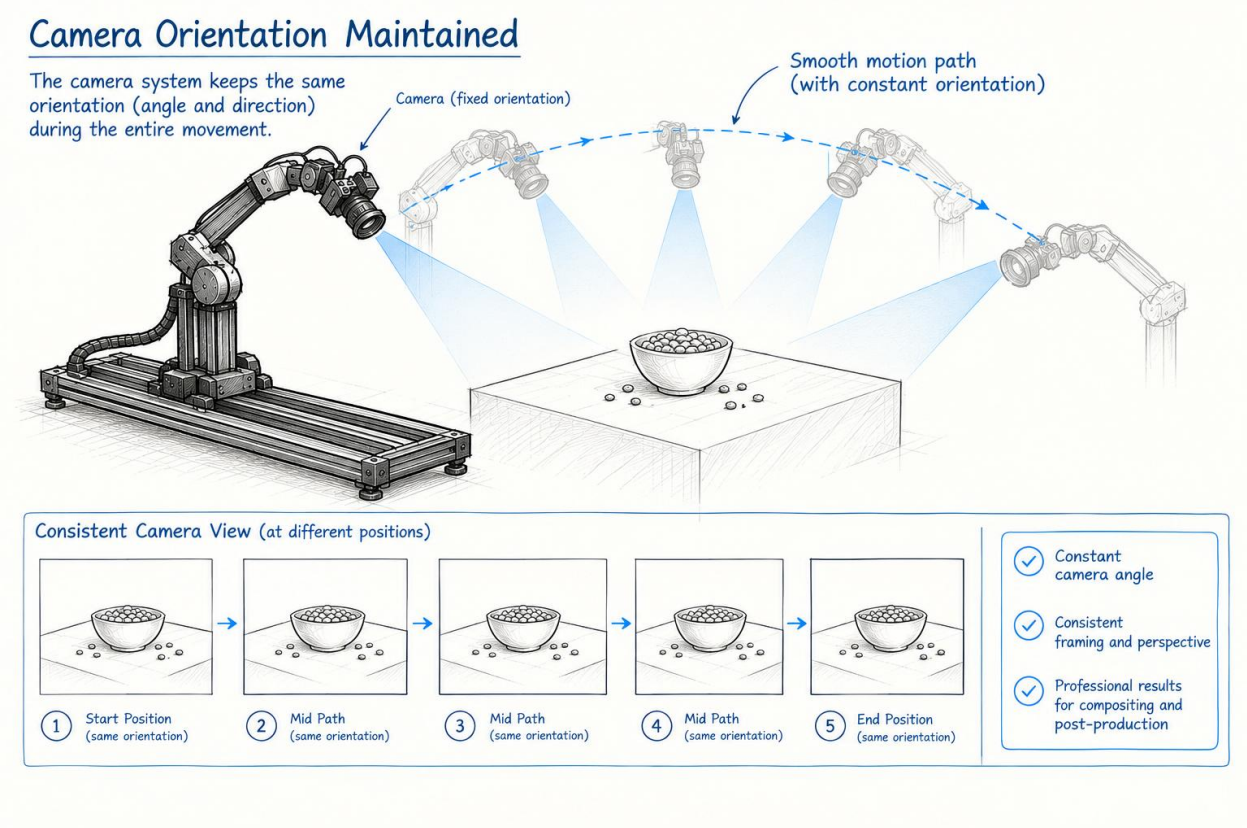
Number 2- Smooth Movement Reproduction

Smooth movement is important for producing usable footage, especially when the camera is moving at higher speeds or when footage is captured at high frame rates. Sudden changes in speed can create visible vibration or unwanted movement in the final footage. Controlled acceleration and deceleration, along with a rigid structure, can help get rid of these effects and provide smoother camera motion. In the long run, this would reduce the number of shots needed and energy wasted.



Number 3 - Camera Orientation

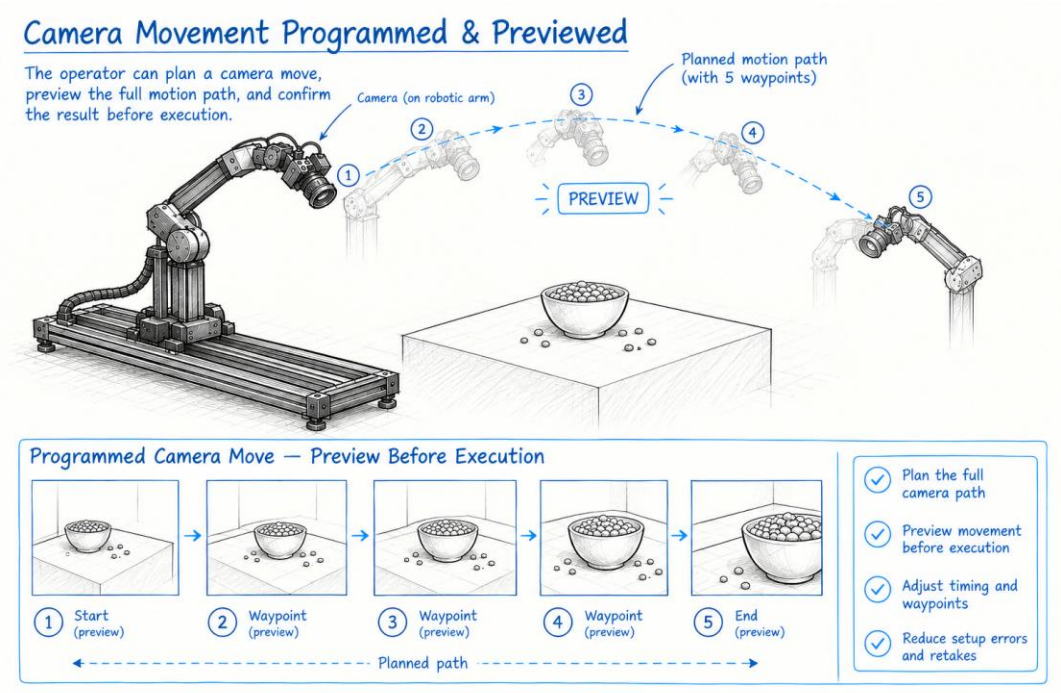
Maintaining the desired orientation is important when the camera is moving around or alongside a product. If the camera angle changes unintentionally during the movement, the object in its vision may not remain framed correctly. So, a motorized rotary or adjustable camera mount could be used with the linear motion system to control the camera's orientation during a programmed movement for taking better shots or keeping the desired angle of shot.





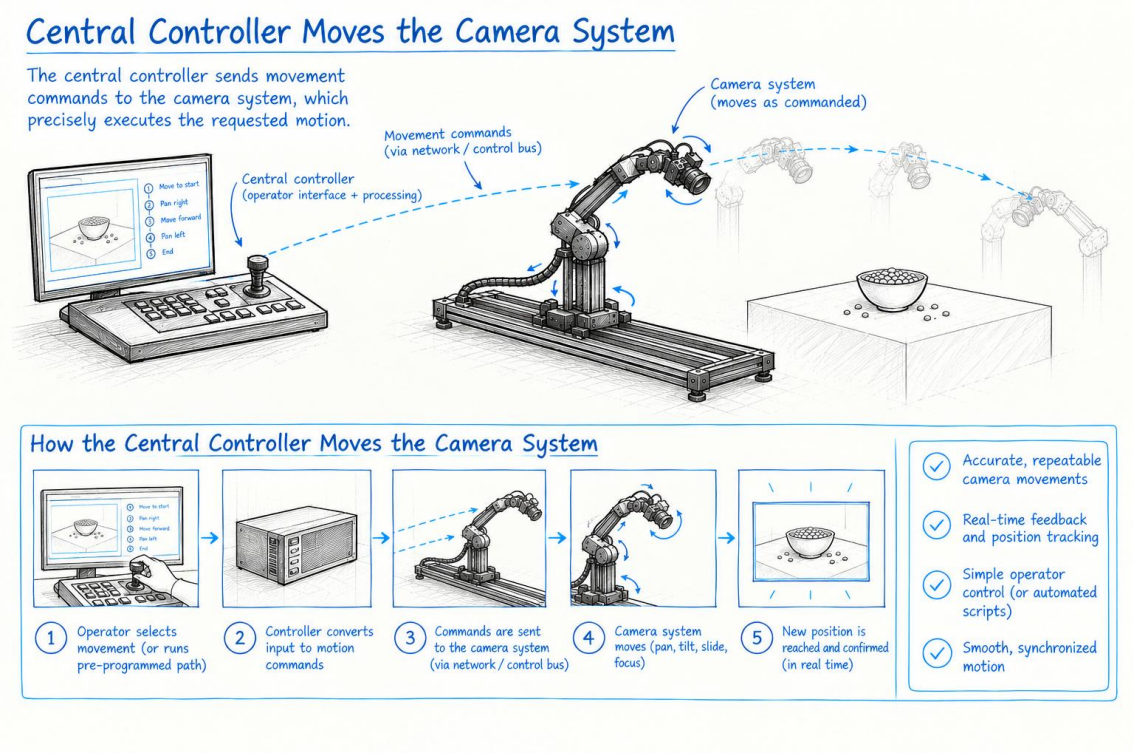
Number 4 - Ease of programming movement

The system should be simple enough that an operator does not need to be able to write code to create a camera movement as the ones using it will mainly be photographers or cinematographers. An interface with a timeline and adjustable movement points could allow the operator to create a programmed path and preview it before the hardware moves, which would decrease the need to have the machine move a lot which can wear it down faster. This could also help identify incorrect movements before running the actual shot and make the system easier to demonstrate.



Number 5 - Centralized Controls

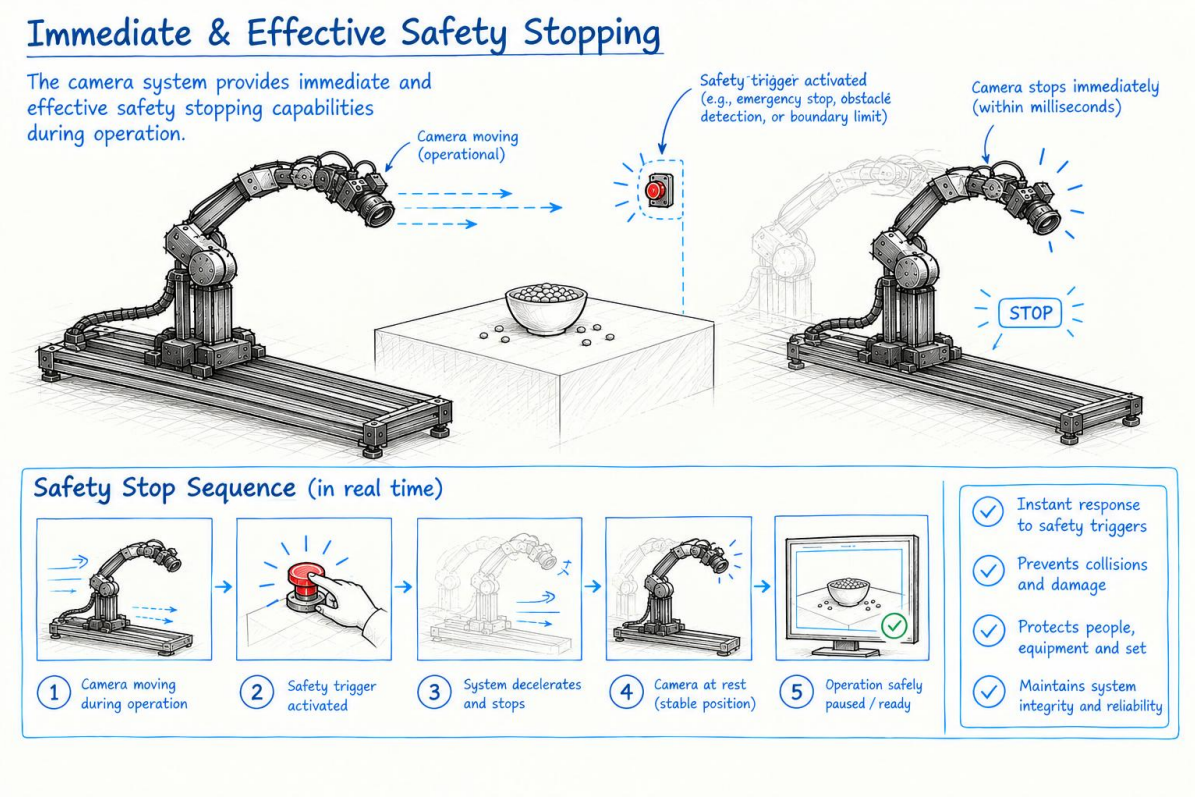
Having different functions controlled from one location would make the system a lot easier to operate. The main controller could be connected to the motors, camera trigger, lighting, and other equipment used during a shot and simplify usage of the machine overall for operators. This would also provide the possibility of synchronizing different systems like lighting, so they operate at the desired time during a programmed movement, reducing the need to move to different parts of the machine for interaction.





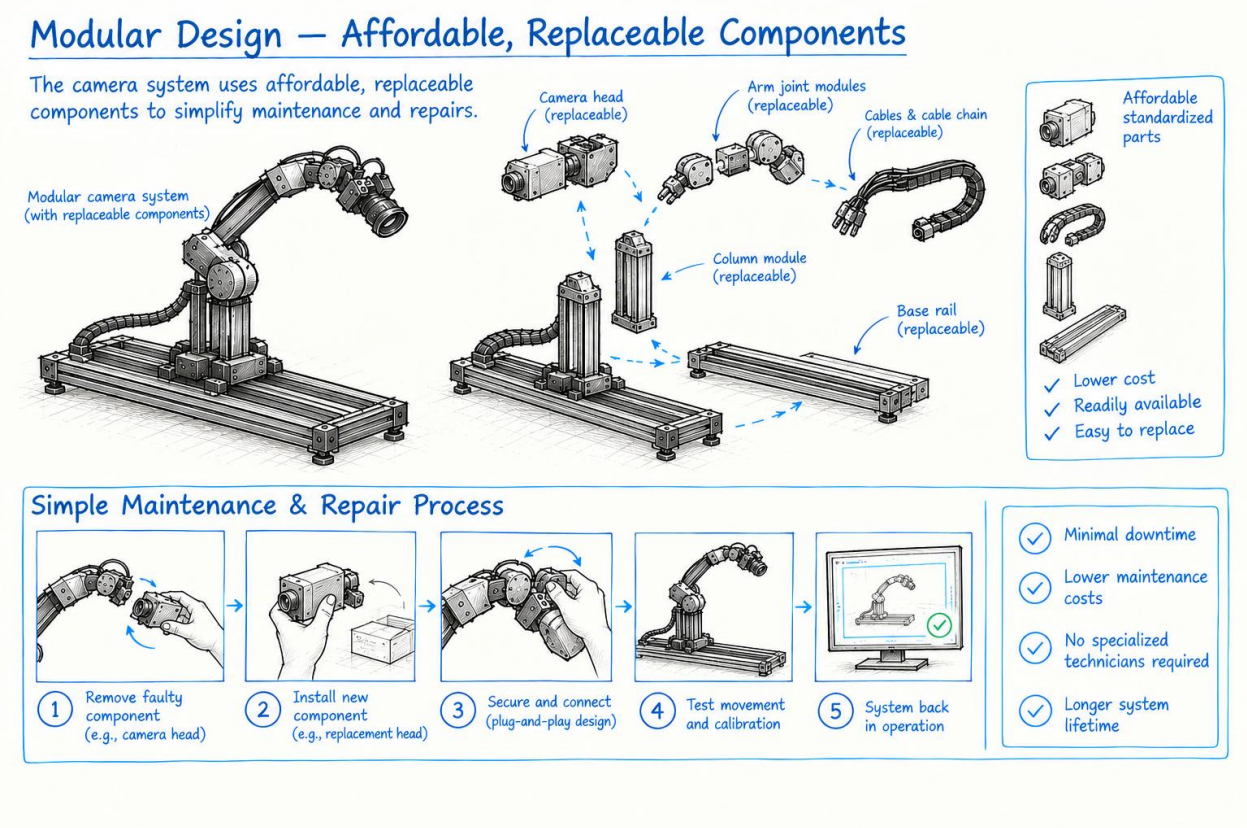
Number 6 - Camera System Safety

Safety is important because the camera system will be operating around people and expensive equipment. An easily accessible emergency stop can allow the operator or another person nearby to stop the system immediately if something goes wrong. With additional features as well such as travel limits and sensors for detecting objects in the movement path would provide further protection for people and equipment. This reduces injuries and costly damage in the long run.



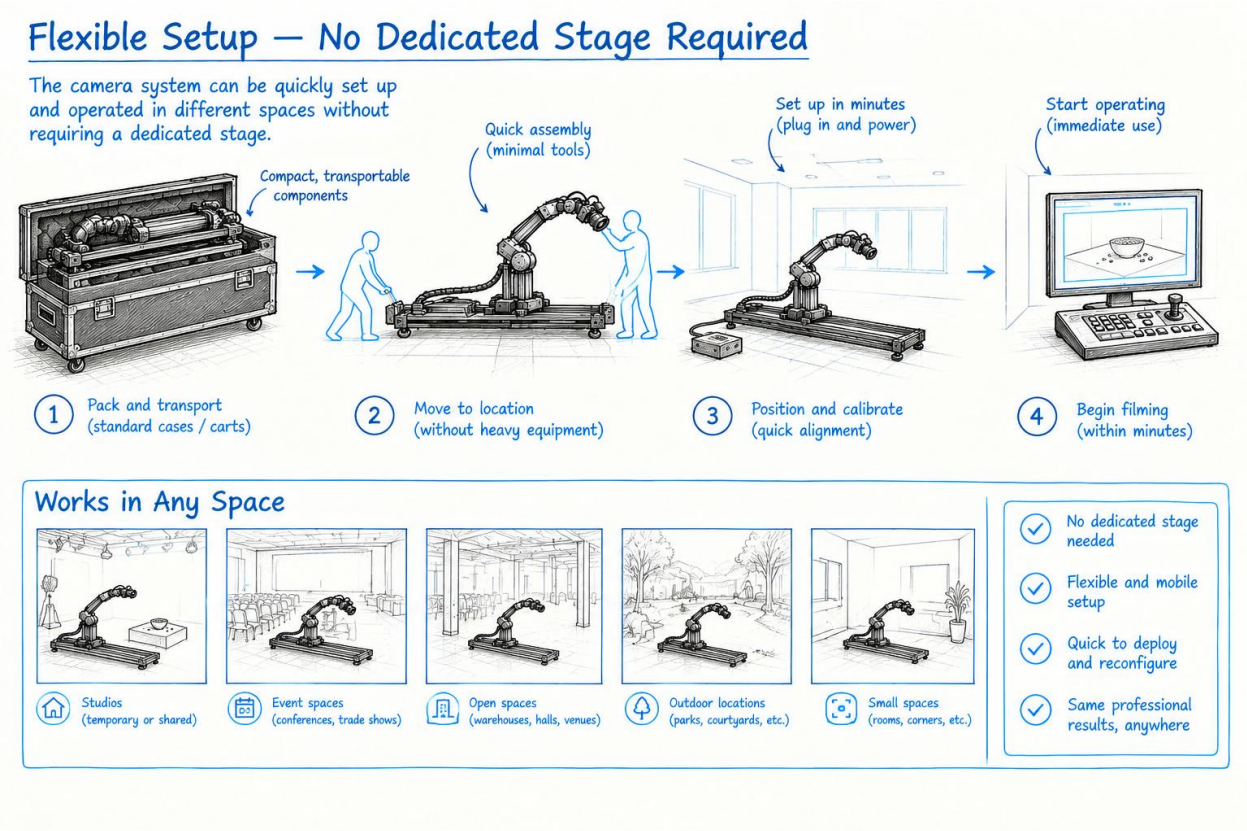
Number 7 – Affordable, Replaceable Components

The camera system is intended to provide a lower-cost alternative to a more expensive robotic camera system. Keeping the system affordable is therefore an important part of the design. Using modular and replaceable components would also make repairs easier as the individual parts could be replaced instead of requiring the entire system to be repaired or replaced. This would help reduce both maintenance costs and downtime.



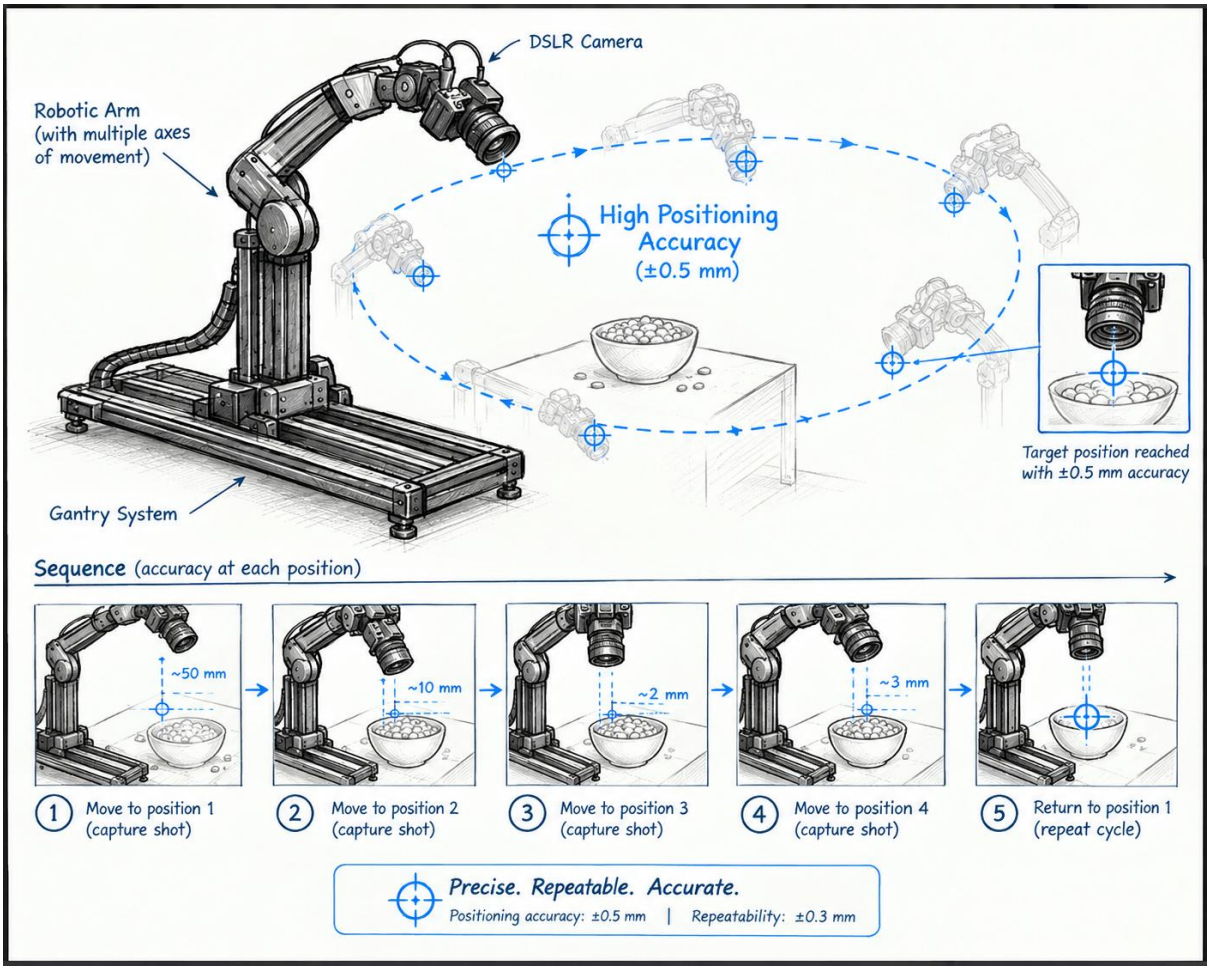
Number 8 - Easy camera system setup

The camera system may be used in classrooms, small studios, and other spaces that are not specifically designed for robotic camera equipment. Because of this, the system should be relatively easy to assemble, move, and pack away. A modular or compact rail system could allow it to be transported between locations and set up without requiring a permanent installation especially since it will be mainly used for shooting product advertisements.



Number 9 – Accuracy on movement

This need is ensuring that each repeated shot matches the needed shot as closely as possible. This is important as differences in camera position, speed, or timing can become noticeable trying to program expected motions. To address this, the system could use position feedback from encoders and a motion controller to compare and reproduce programmed movements. Testing the system by running the same movement multiple times would also help measure how consistently it performs.



Reflection on the Process

Our team has spoken with individuals from the media department in order to gather the necessary information on the feasibility of this project. Some unexpected needs did surface, such as the interest in using the project as a teaching tool more rather than for professional shoots. We watch out for responses in particular around how people go about working on a failed shot and what they would like to have to fix it.

Two areas stand out which is the need for repeatability and maneuverability of camera that cannot be done by hand. A small surprise was how consistently cost came up in the discussion regarding the project rather than a simple preference which led to it being treated as a hard design constraint from the start.

Two changes are planned to improve this process which would be to conduct more interviews from varying backgrounds than just the media faculty and at least one focus group or observation session with students in a media program, which the course notes note can reveal needs faster than one-on-one interviews alone.

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